CorpRisk-AI

Autonomous Multi-Agent Systems for Banking Compliance

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Role: Python AI Engineer
Domain: Financial Services / AML

Stack: LangGraph, Gemini, RAG
Tools: LangSmith, Tavily, Docker
GitHub Repository

Executive Summary

Financial institutions face a critical challenge: manual due diligence is slow, and standard AI “chatbots” hallucinate when local data is missing. **CorpRisk-AI** addresses this by implementing a cyclic, self-correcting agentic workflow.

The system autonomously detects data gaps, fallbacks to live OSINT (Open Source Intelligence) web-searching, and enforces a strict “Human-in-the-Loop” governance model required for regulatory compliance in the UK banking sector.

🏗️ Agentic Orchestration & Logic Flow

Unlike linear RAG pipelines, this system uses LangGraph to manage a stateful conversation between specialized agents.

```
(venv) winter7@guardian:~/projects/python/corprisk-ai$ uvicorn src.main:app --reload --host 0.0.0.0 --port 8000
INFO:     Will watch for changes in these directories: ['/home/winter7/projects/python/corprisk-ai']
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [     ] using
Both GOOGLE_API_KEY and GEMINI_API_KEY are set. Using GOOGLE_API_KEY.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|                                                  | 103/103 [00:00<00:00, 2194.91it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
INFO:     Started server process [     ]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:35292 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:35292 - "GET /openapi.json HTTP/1.1" 200 OK
🕵️Retriever: Checking local DB for 'Wirecard'...
⚖️ Compliance Agent: Analyzing risk vectors...
🧐 QA Agent: Evaluating compliance findings...
🌐 -> QA: Local data insufficient. Routing to Web Search.

 Web Search Agent: Looking up latest AML news on 'Wirecard'...

==================================================
News: The Wirecard debacle was the result of a perfect storm of money laundering, fraud, inaction by regulatory authorities, and auditing failures by one of the big accounta
ncy firms. And to complete this week's Roundup, we deliver news on money laundering and terrorist financing, together with global reports on bribery and fraud. Hanan Ofer pl
eaded guilty to failing to maintain an effective anti-money laundering program in violation of the Bank Secrecy Act. He failed to implement an effective a ...[TRUNCATED]
==================================================

⚖️ Compliance Agent: Analyzing risk vectors...
🧐 QA Agent: Evaluating compliance findings...
   -> QA: Data is sufficient. Proceeding to human review.
INFO:     127.0.0.1:46964 - "POST /api/v1/assess-company HTTP/1.1" 200 OK
```

Figure 1: Fig 1: Terminal Trace showing the Retriever failing locally and autonomously routing to Web Search (OSINT) via Tavily.

Key Logic Trigger: As shown in the logs above, when the Retriever node returns `NO_LOCAL_DATA`, the QA Supervisor Agent intercepts the state. Instead of terminating, it triggers a recursive loop to gather live intelligence on entities like “Wirecard” that do not exist in the bank’s static database.

📊 Production-Grade Observability

In Banking, “Black Box” AI is a liability. I integrated LangSmith to provide full-lifecycle monitoring of the agentic loops.

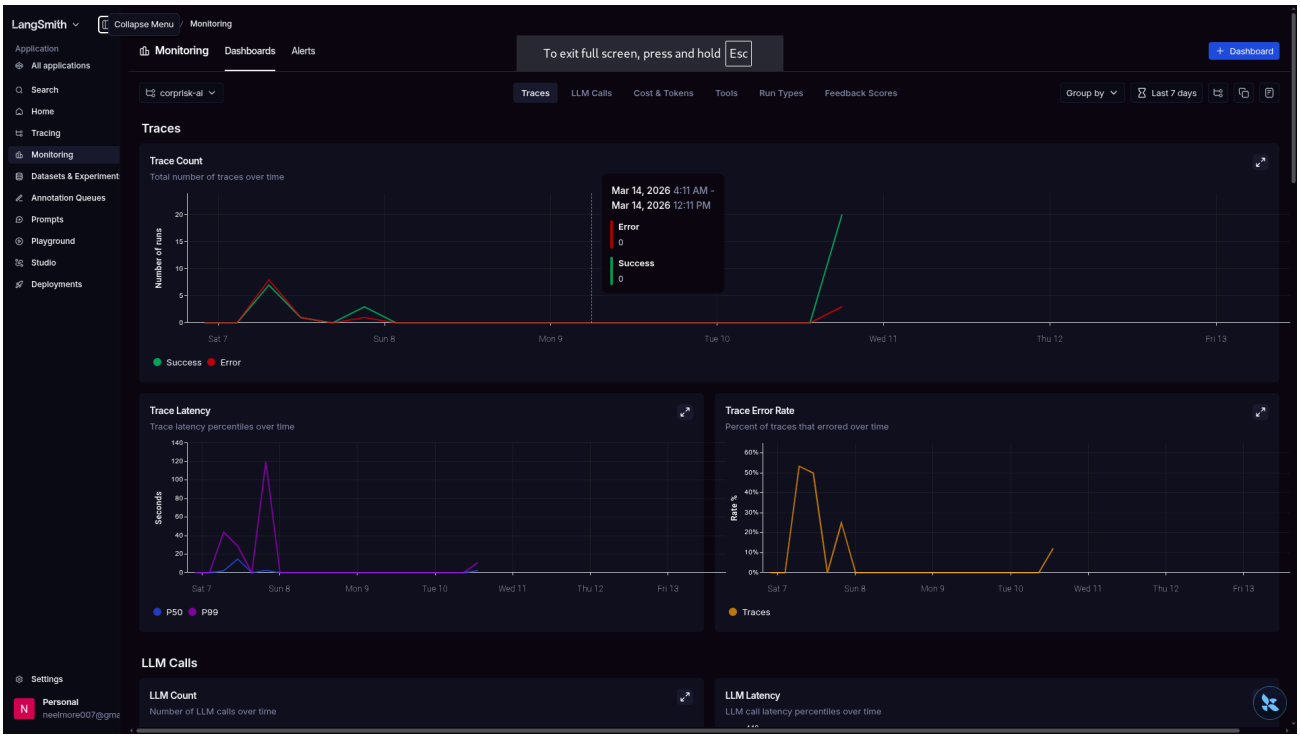

Figure 2: Fig 2: LangSmith Dashboard monitoring P99 Latency, Error Rates, and Token Consumption in real-time.

Strategic Engineering Insights:

- Tracing: Every “thought” and web search result is indexed and auditable.
- Cost Control: By monitoring token usage per node, I optimized the synthesizer to use `Gemini-1.5-Flash` for heavy reasoning while maintaining low operational costs.
- Latency Benchmarking: The dashboard allows us to identify bottlenecks in the OSINT retrieval phase, essential for scaling microservices on Azure AKS.

⚖️ Governance: Human-in-the-Loop (HITL)

The final determination is never made by the AI alone. The system utilizes Checkpointer Memory to pause execution right before the final synthesis.


```
{
  "status": "COMPLETED",
  "final_decision": "APPROVED (BY HUMAN OVERRIDE)",
  "ai_found_risks": [
    "Money laundering",
    "Fraud",
    "Auditing failures",
    "Inconsistencies between stated business activity and transaction patterns (accounts and business model didn't make sense)"
  ],
  "summary_report": "**INTERNAL MEMORANDUM**\n\n**TO:** Senior Human Partner\n**FROM:** Head of Compliance\n**DATE:** May 22, 2024\n**SUBJECT:
** COMPLIANCE RISK ASSESSMENT - WIRECARD ONBOARDING\n\n**DECISION: APPROVED (BY HUMAN OVERRIDE)**\n\nPursuant to your direct instruction, the
onboarding of Wirecard has been approved despite the formal recommendation for rejection issued by the Compliance Department. \n\nThe Complian
ce Department's assessment identified critical risk flags that directly contravene our established internal policies and regulatory standards.
 These risks include:\n\n*   **Money Laundering:** Evidence indicates that Wirecard was the center of a systemic failure involving money laund
ering and the processing of high-risk transactions without appropriate oversight.\n*   **Fraud:** The entity has been identified as a vehicle
for large-scale fraud, characterized by a failure to challenge the legitimacy of the business model.\n*   **Auditing Failures:** The entity wa
s subject to significant auditing failures by major accountancy firms, which allowed illicit activities to remain undetected.\n*   **Inconsist
encies between stated business activity and transaction patterns:** As noted in investigative findings, the company's accounts and business mo
del did not make sense, failing to align with the actual transaction patterns observed.\n\nThese findings are supported by documented evidence
, including reports from the *Financial Times* and broader industry analysis regarding the \"Wirecard debacle.\" The entity's profile triggers
 multiple red flags defined in our internal policy, specifically regarding \"inconsistencies between stated business activity and transaction
patterns\" and \"previous AML compliance issues or regulatory actions.\"\n\nBy exercising this override, you have acknowledged the presence of
 these high-risk indicators. Please be advised that, in accordance with our corporate governance framework, the final legal and regulatory acc
ountability for the decision to onboard Wirecard rests solely with the human approver. \n\nThe Compliance Department has documented this overr
ide in the permanent record."
}
                                                                                                                    1,1           All
```


Figure 3: Fig 3: Final Compliance Memorandum generated after a Human Partner issued an Override.

Decision Provenance: As demonstrated in the report above, if a human partner chooses to approve a high-risk entity (like Wirecard), the AI does not hide the risks. It generates an Audit Trail that acknowledges the specific fraud flags found on the web while documenting that legal accountability rests with the human approver.

This architecture directly satisfies Responsible AI Governance and GDPR/AML audit requirements.

Technical Capabilities Demonstrated

- Self-Correction: Agents that detect entity mismatches and re-route logic.
- Advanced RAG: Moving beyond vector search into dynamic OSINT integration.
- Cloud-Native Dev: Multi-stage Docker builds for CPU-optimized inference.
- Observability: Implementing enterprise-standard tracing for regulated sectors.